

Lightweight AI Image Detection: CNN vs. Transformer at Consumer Scale

Shea Stevens

University of Hawai‘i at Mānoa
ICS 605 Applied AI, May 2026

Abstract

AI-generated images have become increasingly difficult to distinguish from real photographs, yet robust detection systems require large-scale compute and training regimes inaccessible to small teams. This work investigates whether lightweight, consumer-trainable models can detect AI-generated images across unseen generators, and whether a CNN or a vision transformer is better suited to that task under practical constraints. We fine-tune EfficientNet-B0 and DeiT-Tiny under two conditions: a constrained setting using five generator types (~25,000 images) and a broad setting using all ArtiFact generators (~165,000 images). Under constrained training, DeiT-Tiny achieves 63.5% cross-generator accuracy versus 60.7% for EfficientNet-B0, with both models failing severely on real portrait photography due to training data composition. Under broad training, accuracy rises to 88.7% (EfficientNet-B0), 87.6% (DeiT-Tiny), and 91.2% (ensemble) and eliminates portrait bias. Both models fail on external commercial generators (Midjourney, Flux, DALL-E 3, SDXL), dropping to 27–39% overall. This failure suggests the models learn pipeline-specific artifacts rather than general properties of synthetic images, and that detection accuracy does not transfer across generation paradigm shifts. We package both models into a Firefox browser extension using ONNX Runtime Web, enabling on-device inference with no server dependency.

1. Introduction

The rapid advancement of generative models has produced images routinely indistinguishable from real photographs. The downstream implications span misinformation, synthetic identity fraud, and the erosion of digital trust.

Prior detection work has focused on large-scale, high-resource settings with models containing hundreds of millions of parameters and training sets exceeding 100,000 images. This framing is ill-suited for small teams and fails to stay stable and accurate amongst the increasingly diverse generator landscape, further exacerbating large-scale training pre-requisites.

This project constrains itself to consumer-accessible conditions: both models have under 6 million parameters and training runs on a single GPU session in approximately 8 hours. We evaluate two training configurations, a limited five-generator setting and a broad all-generator setting, then test both against commercial generators entirely unseen during training. The central question is whether a CNN or transformer generalizes better under these constraints, and whether that generalization holds beyond the training distribution.

We also contribute a browser extension that runs detection entirely on the user’s device, eliminating privacy concerns associated with server-side detection APIs.

Contributions:

1. A systematic comparison of EfficientNet-B0 and DeiT-Tiny for cross-generator AI image detection under consumer-scale constraints
2. A per-generator accuracy analysis across 26 unseen generators identifying specific failure modes
3. An out-of-distribution evaluation against four contemporary commercial generators, establishing the limits of ArtiFact-trained detectors
4. An open-source Firefox browser extension providing on-device detection with no network dependency
5. A Grad-CAM interpretability analysis showing how the constrained-to-broad training shift changes what each architecture attends to, providing visual, mechanistic evidence for the portrait-bias finding and for why ensembling only helps under broad training

2. Background

AI image detection has largely been studied as a high-resource problem. The dominant approach fine-tunes large pretrained vision models (ResNet-50, ViT-Base, CLIP) on datasets with hundreds of thousands of images spanning many generator types, producing strong within-distribution accuracy. The weakness of this approach is that the detection signal tends to be generator-specific: models trained on GANs do not reliably transfer to diffusion models, and models trained on early diffusion systems do not transfer to flow-matching architectures.

The generator landscape has shifted substantially. Early GAN-based systems (ProGAN, StyleGAN) leave characteristic frequency artifacts in the Fourier domain that relatively shallow classifiers can detect. Latent diffusion models (Stable Diffusion, DALL-E 2) produce subtler artifacts tied to the denoising schedule. Contemporary systems such as Midjourney v6, Flux, and DALL-E 3 apply improved schedulers and photorealistic fine-tuning that largely suppress these earlier artifact signatures.

Browser-based on-device detection has received less attention. Deploying models as ONNX binaries via WebAssembly avoids server round-trips and the associated privacy risks, but requires models small enough to transfer and execute within browser memory constraints. EfficientNet-B0 and DeiT-Tiny are among the smallest architectures that retain competitive accuracy on ImageNet-scale tasks, making them natural candidates for this setting.

This project sits at the intersection of these two directions: small-model detection evaluated against the full generator diversity spectrum, culminating in a browser-deployable system.

3. Data

We use the **ArtiFact** dataset, which contains images from 33 generative sources and several real photography collections, each with a metadata file recording generator labels.

Constrained training (Section 5). Five sources: `pro_gan`, `stable_diffusion`, `stylegan1`, `latent_diffusion`, and `coco`. The first four span GANs and early diffusion models; `coco` provides the real-image signal (natural scene photography only). Each source is capped at 5,000 images, yielding 25,000 samples split 90/10 train/validation.

Broad training (Section 6). All ArtiFact generator folders, approximately 165,000 images total with up to 5,000 images per generator. 500 images per folder are withheld as a stratified holdout (16,500 total).

Evaluation. For the constrained condition, all 26 generator folders unused in training serve as the held-out evaluation set (500 images each, 13,000 total). This includes unseen GANs, diffusion models, and real photography collections. The stratified holdout is used for the broad training model. Notably, most real-photography evaluation sources consist primarily of portrait and face images, a domain absent from the constrained training data.

4. Method

4.1. Model Architectures

We fine-tune two pretrained architectures from the `timm` library, matched in parameter count and representing distinct inductive biases.

Table 1. Cross-generator metrics, constrained training.

Model	Acc	Prec	Rec	F1	AUC
EfficientNet-B0	60.7	73.1	71.0	72.0	0.556
DeiT-Tiny	63.5	74.3	74.7	74.5	0.597
Ensemble	63.1	74.0	74.4	74.2	0.581

EfficientNet-B0 is a convolutional architecture with inverted residual blocks operating on local receptive fields, making it sensitive to local texture artifacts. The model has approximately 5.3 million parameters and exports to a 16 MB ONNX binary.

DeiT-Tiny is a data-efficient vision transformer operating on 16×16 patches. Its multi-head self-attention aggregates information globally, potentially capturing broader structural regularities of AI generation. It has approximately 5.7 million parameters and exports to a 22 MB ONNX binary.

Both use a 2-class head (real vs. AI), pretrained ImageNet weights, and are fine-tuned end-to-end.

4.2. Training

Both models are trained identically: AdamW ($\text{lr}=3\text{e-}5$ for constrained training, $\text{lr}=1\text{e-}4$ for broad training, $\text{weight_decay}=1\text{e-}4$), cosine annealing, cross-entropy loss, and early stopping with $\text{patience}=4$. Augmentations include random horizontal flip and color jitter (brightness and contrast ± 0.2). All images are resized to 224×224 and normalized with ImageNet statistics.

Training ran on a single NVIDIA T4 GPU (Google Colab). Both models converged within 20 epochs. Wall-clock time was approximately one to two hours per model for the constrained setting. The broad setting required multiple GPU-hours per epoch, approximately 8 hours for total training.

4.3. ONNX Export and Browser Deployment

Checkpoints are exported to ONNX opset 18 using `torch.onnx.export` with constant folding. The Firefox extension loads both models at startup via ONNX Runtime Web 1.18.0 (WebAssembly, single-threaded to satisfy browser security constraints) and holds them in memory for zero-latency inference. The JavaScript preprocessing pipeline exactly mirrors the Python training pipeline; we verified logit equivalence on held-out samples to rule out preprocessing mismatch.

5. Experiments: Constrained Training

5.1. Overall Results

Table 1 shows cross-generator metrics on 13,000 held-out images (26 generators $\times$ 500 images).

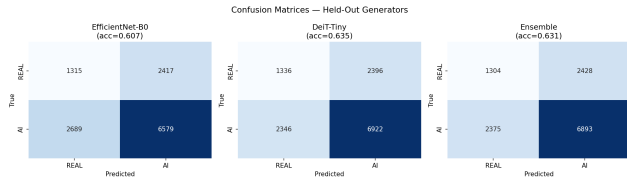


Figure 1. Confusion matrices for EfficientNet-B0 and DeiT-Tiny, constrained training.

Table 2. Per-generator accuracy, selected generators (constrained training).

Generator	Type	Eff-B0	DeiT
denoising_diffusion_gan	AI	99.6	100.0
sfhq	AI	99.8	100.0
star_gan	AI	96.8	98.6
generative_inpainting	AI	21.4	14.0
gau_gan	AI	42.2	55.6
imagenet	Real	61.6	70.0
celebahq	Real	3.8	1.8
ffhq	Real	2.4	1.6

DeiT-Tiny outperforms EfficientNet-B0 across all metrics. The ensemble offers no improvement; both models fail on the same generators, suggesting correlated errors rather than complementary failure modes.

The 60–63% headline figures are substantially suppressed by a single cause: real portrait photography. Excluding the four portrait-heavy real folders (ffhq, celebahq, afhq, metfaces), accuracy rises to 69.5% (EfficientNet-B0) and 73.4% (DeiT-Tiny), an improvement of nearly 10 points. The low headline numbers reflect a specific, correctable training data gap.

5.2. Per-Generator Analysis

Table 2 shows accuracy on selected generators. Models perform extremely well on generators with strong frequency or texture artifacts, and fail on face-centric generators and real portrait photography.

5.3. Portrait-Domain False-Positive Bias

FFHQ achieves only 2.4% accuracy under EfficientNet-B0, meaning 97.6% of real faces are flagged as AI. This is attributable to a training data gap: the only real-image source in constrained training was coco, which contains natural scenes but no face photography. High-quality portrait photography shares visual characteristics with GAN face outputs, such as smooth skin rendering, controlled lighting, and centered composition, and without real portrait examples neither model can separate genuine faces from synthetic ones.

This is not an architectural failure; it is a data composition problem that broad training directly addresses.

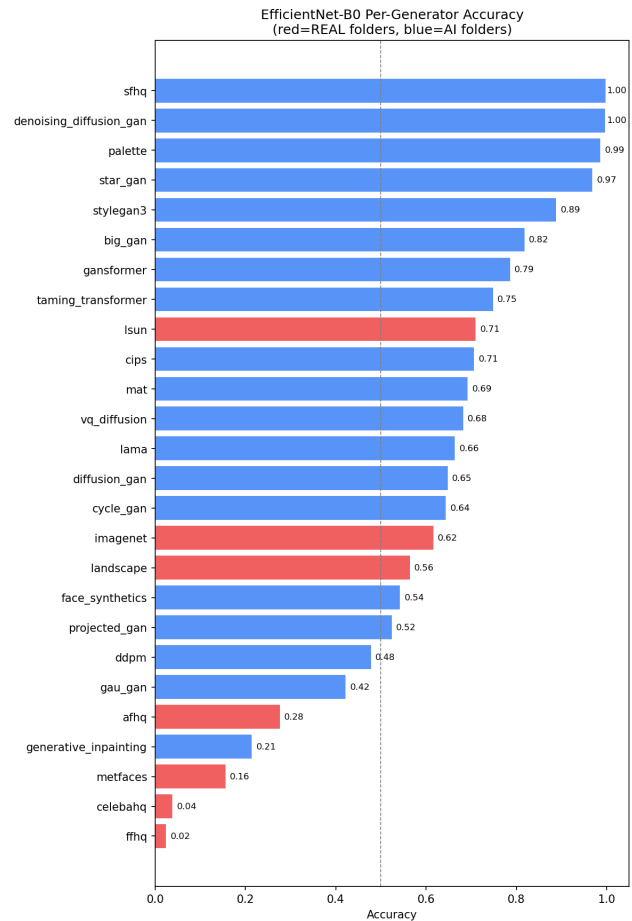


Figure 2. Per-generator accuracy across all 26 held-out generators, constrained training.

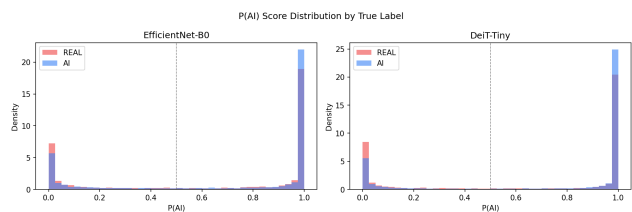


Figure 3. Confidence score distributions by class, constrained training.

5.4. Inference Timing

Single-image inference via ONNX Runtime Web on CPU (WebAssembly) is sub-100 ms for both models, making the extension responsive for interactive use. EfficientNet-B0 is faster due to its smaller ONNX binary (16 MB vs. 22 MB for DeiT-Tiny).

Table 3. Constrained vs. broad training.

Model	Constrained	Broad	Δ	AUC (Broad)
EfficientNet-B0	60.7	88.7	+28.0	0.960
DeiT-Tiny	63.5	87.6	+24.1	0.937
Ensemble	63.1	91.2	+28.1	0.964

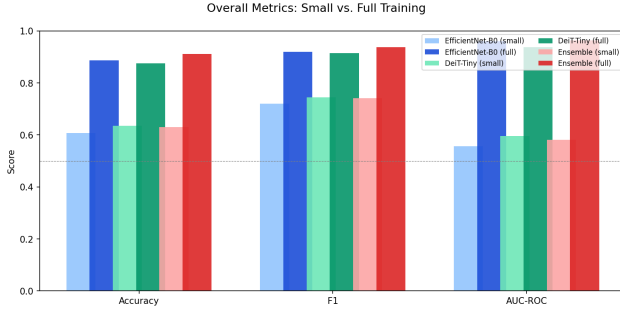


Figure 4. Overall metrics comparison: constrained vs. broad training.

6. Experiments: Broad Training

To directly test whether training data breadth drives generalization, we trained both models on all ArtiFact generators with a 500-image stratified holdout per folder.

6.1. Results

Table 3 shows constrained vs. broad accuracy. Broad training yields a 28-point accuracy gain, establishing generator diversity as the dominant factor in cross-generator generalization. Three additional findings emerge.

Architecture ranking reverses. DeiT-Tiny led under constrained training; EfficientNet-B0 leads under broad training (88.7% vs. 87.6%, with a larger advantage in AUC-ROC). EfficientNet’s local texture sensitivity appears better matched to GAN artifact detection when it has sufficient training diversity to leverage it.

Ensemble becomes effective. Under constrained training the ensemble offered no improvement, indicating both models failed on the same inputs. Under broad training the ensemble gains 2.5 points over the best single model, indicating the two architectures now fail on complementary subsets.

Portrait bias resolved. FFHQ accuracy rises from 2.4% to 91.6%; CelebA-HQ from 3.8% to 96.2%; AFHQ from 27.6% to 95.4%. The one remaining underperformer (MetFaces, 72.4%) consists of historical portrait paintings, not photographs.

Even under broad training, image completion models remain hard: `generative_inpainting` (65.4%) and `stylegan2` (66.2%).

7. Interpretability: What Are the Models Attending To?

Sections 5.3 and 6.1 establish *that* broad training resolves the portrait-domain false-positive bias; this section provides visual evidence for *why*, addressing reviewer feedback asking for insight into what the models actually look at when making a real/AI decision.

7.1. Method

We use Grad-CAM (Selvaraju et al., 2017) to visualize the image regions driving each model’s predicted class. Target layers follow architecture-appropriate convention: the final convolutional activation before global pooling for EfficientNet-B0, and the first LayerNorm of the last transformer block for DeiT-Tiny (the standard recipe for applying Grad-CAM to a ViT, reshaping the 196 patch tokens back into a 14×14 spatial grid and discarding the CLS token).

Neither the constrained nor the broad checkpoint exists as a .pt file locally — both were trained on Colab and only their ONNX exports (`*best_better_settings.onnx` and `*best_full.onnx`, the same weights shipped in the browser extension) survived. Since Grad-CAM needs gradients, which a standard ONNX Runtime inference session cannot provide, we reconstructed both into differentiable PyTorch modules via `onnx2torch`, patching two opset-18 operator conversions the library did not yet support (`ReduceMean` with `axes-as-input`, `Reshape` with `allowzero=1`). We verified the reconstructed modules reproduce the original ONNX Runtime output to within 2e-5 across all sample images before trusting any heatmap derived from them.

7.2. Results

Figure 5 shows both architectures’ Grad-CAM output on the same real FFHQ portrait under both training conditions. Under constrained training, both models predict AI for this photo; under broad training, both correctly predict Real — the same flip responsible for FFHQ’s 2.4%-to-91.6% accuracy swing (Section 6), now made visible.

The two architectures attend to the image differently, consistent with their differing inductive biases (Section 4.1). EfficientNet-B0 produces one tight, concentrated hotspot centered on the nose and mouth — consistent with a convolutional stack keying off a single dominant local-texture cue (under constrained training, apparently mistaking natural skin/lighting detail for a synthesis artifact). DeiT-Tiny’s attention is diffuse and blocky, following its 14×14 patch grid and spreading across the eyes, hairline, and background simultaneously, consistent with self-attention mixing multiple discrete regions rather than committing to one.

This divergence also gives a mechanistic account of the ensemble finding in Section 6: under constrained training

both architectures fail on the same inputs because both are reacting to the same missing-data gap (no real portraits in the training set), so averaging two correlated failures gains nothing. Under broad training, the two models’ different attention patterns mean they are keyed off different visual evidence for the same decision — which is precisely why their errors become complementary rather than correlated, and why ensembling them yields a real 2.5-point gain rather than none.

7.3. Control: Is This a Real Shift or a Target-Class Artifact?

Because Figure 5 targets each model’s *predicted* class, and the constrained and broad models predict opposite classes for this photo (AI vs. Real), part of the apparent shift could simply reflect that we are visualizing evidence for two different logits, not the same detector before and after training — on a shared backbone with a 2-way linear head, “evidence for AI” and “evidence for Real” are not required to occupy the same pixels. To isolate this, we recompute Grad-CAM for all four models with the target class fixed to AI, regardless of what each model actually predicts (Figure 6).

The relocation persists under this control, ruling out a pure target-class artifact. For EfficientNet-B0, the concentrated nose/mouth hotspot that dominates the constrained model’s AI-evidence all but disappears under broad training: the strongest remaining activation is far weaker and shifts to the image periphery (hair/corner), away from the central face entirely. This suggests broad training did not merely reweight the same central-face cue toward “real” — it stopped treating that region as diagnostic of synthesis. For DeiT-Tiny, the AI-evidence stays strong under both conditions but redistributes, from one isolated cheek/jaw region under constrained training to a broader, multi-patch pattern spanning the eyes and hairline under broad training.

Grad-CAM’s per-image normalization means these maps are only comparable in relative spatial layout, not absolute magnitude across models; we do not read EfficientNet-B0’s weaker broad-training activation as evidence the model is “less certain,” only that the location of whatever AI-evidence exists has moved. Taken together, the decision flip (Figure 5) and this class-agnostic control (Figure 6) point to the same conclusion: broad training changes which visual cues each architecture treats as informative, not merely how it weighs a fixed set of cues.

8. Out-of-Distribution Evaluation

Sections 5 and 6 evaluate generalization within ArtiFact; Section 7 visualizes what drives it. To test whether that performance holds against generators entirely absent from training, we collected 225 images from six contemporary commercial generators and one real-photography source (Table 4).

Table 4. External OOD evaluation sources.

Source	Type	N
Midjourney (v6)	AI	40
Flux	AI	23
SDXL (Civitai)	AI	40
DALL-E 3	AI	23
GPT Images 2	AI	40
Gemini (image gen.)	AI	19
Picsum (real photos)	Real	40

Table 5. External OOD accuracy (all models). “Better” = constrained training, “Full” = broad training.

Model	Acc	Prec	Rec	F1
EfficientNet-Better	37.8	83.6	30.3	44.4
DeiT-Better	24.9	71.1	14.6	24.2
EfficientNet-Full	26.2	100	10.3	18.6
DeiT-Full	29.8	100	14.6	25.5
Ensemble-Better	30.7	82.2	20.0	32.2
Ensemble-Full	25.3	100	9.2	16.8

Table 6. Per-source accuracy, selected models.

Source	Eff-Better	Eff-Full
DALL-E 3	43.5	4.3
Flux	34.8	26.1
Midjourney	27.5	10.0
SDXL	17.5	0.0
GPT Images 2	27.5	2.5
Gemini	47.4	36.8
Real photos	72.5	100.0

All six AI sources use latent diffusion, flow-matching, or transformer-based architectures, representing a different generation paradigm from the GAN- and early-diffusion-heavy ArtiFact distribution.

8.1. Results

Overall accuracy falls to 25–38% compared to 88–91% on ArtiFact, a roughly 55-point degradation. The full models show extreme conservatism: 100% precision with 9–15% recall, meaning they almost never predict AI on external images. Real photo accuracy is perfect (100%) while AI source accuracy collapses, with SDXL reaching 0% and GPT Images 2 reaching 2.5% under the full models.

8.2. Accuracy Degradation When Generalizing

The uniformity of the degradation indicates the models are not learning a general representation of synthetic content. They have learned pipeline-specific signatures characteristic of ArtiFact generators that are absent in contemporary commercial output.

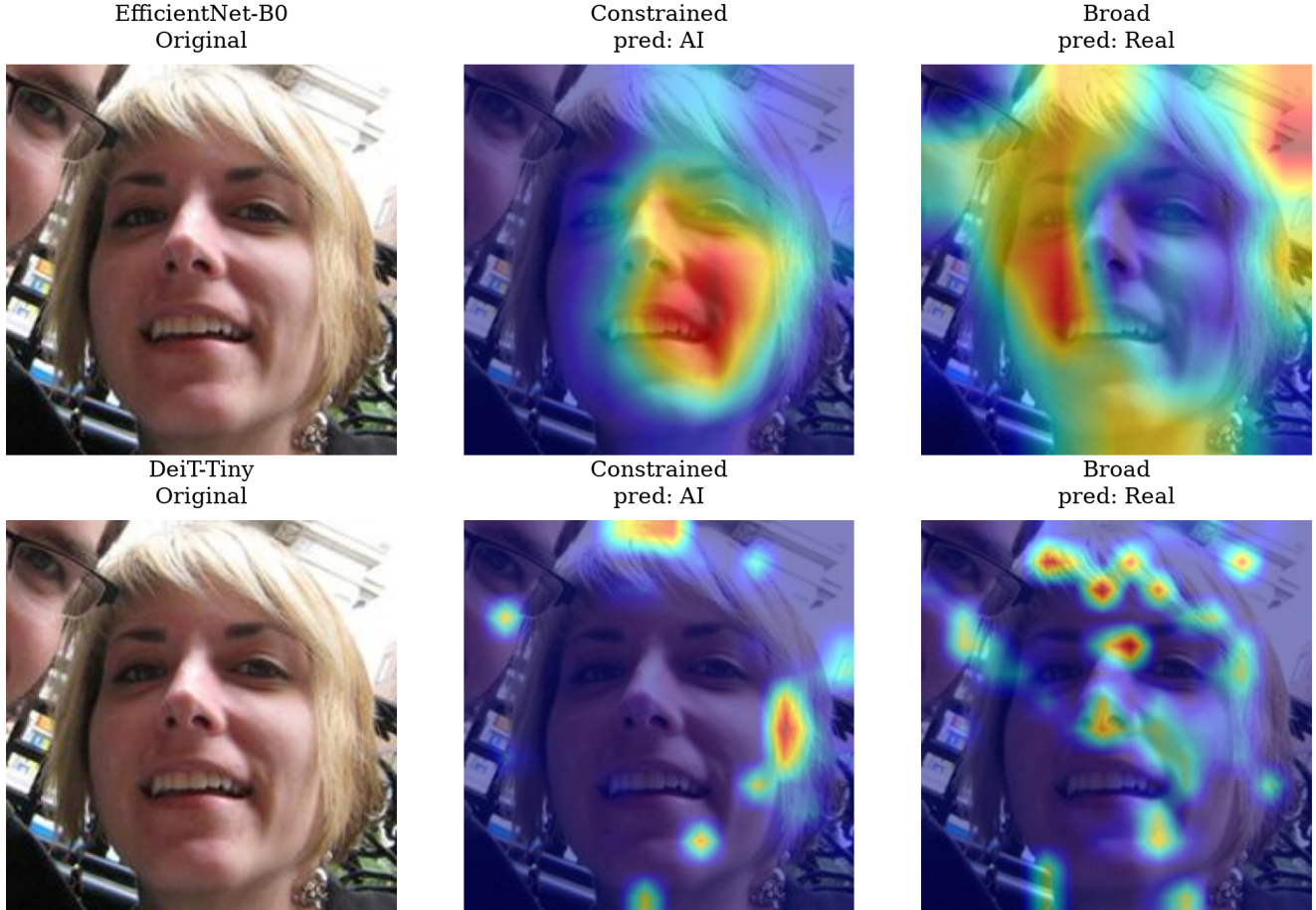


Figure 5. Grad-CAM attention on the same real FFHQ portrait, constrained vs. broad training, for both architectures. Both models flip from misclassifying the photo as AI to correctly classifying it as Real. EfficientNet-B0’s attention concentrates on one local region (nose/mouth); DeiT-Tiny’s spreads across multiple patch-grid regions.

Two compounding factors explain this. First, ArtiFact skews toward GANs and early diffusion models with well-documented frequency artifacts. Midjourney v6 and Flux use flow-matching and improved schedulers that produce cleaner outputs with fewer of these artifacts. Second, the external generators are among the most photorealistic systems available; the models were trained on data where highly realistic outputs were real photographs, so photorealistic AI output activates the same features as real photos.

The core implication is that data coverage is paramount. Broad training dramatically improves generalization within a known generator landscape, but that advantage does not transfer across generation paradigm shifts. Detection models learn generator-family-specific identifiers, and those identifiers become obsolete as the field advances.

9. Conclusion

Lightweight models under consumer-scale constraints can achieve strong cross-generator detection accuracy when

trained on a sufficiently diverse generator set, reaching 88–91% on held-out ArtiFact generators. Training data diversity is the dominant factor: a 28-point gain from constrained to broad training outweighs any architectural difference. The architecture ranking itself is data-dependent: DeiT-Tiny leads under limited training, EfficientNet-B0 leads under broad training, and the ensemble only becomes beneficial once the two models fail on complementary subsets.

However, accuracy does not hold against generators from a different generation paradigm. Both models collapse to 27–39% on Midjourney, Flux, SDXL, and DALL-E 3, suggesting they learn pipeline-specific artifacts rather than any invariant property of synthetic images. The practical implication is that a small team with sufficient data access can build an accurate detector for today’s landscape, but that accuracy is not guaranteed to persist as generation methods advance. Continuous retraining against newly released generators appears necessary to maintain useful performance.

The browser extension deployment demonstrates that

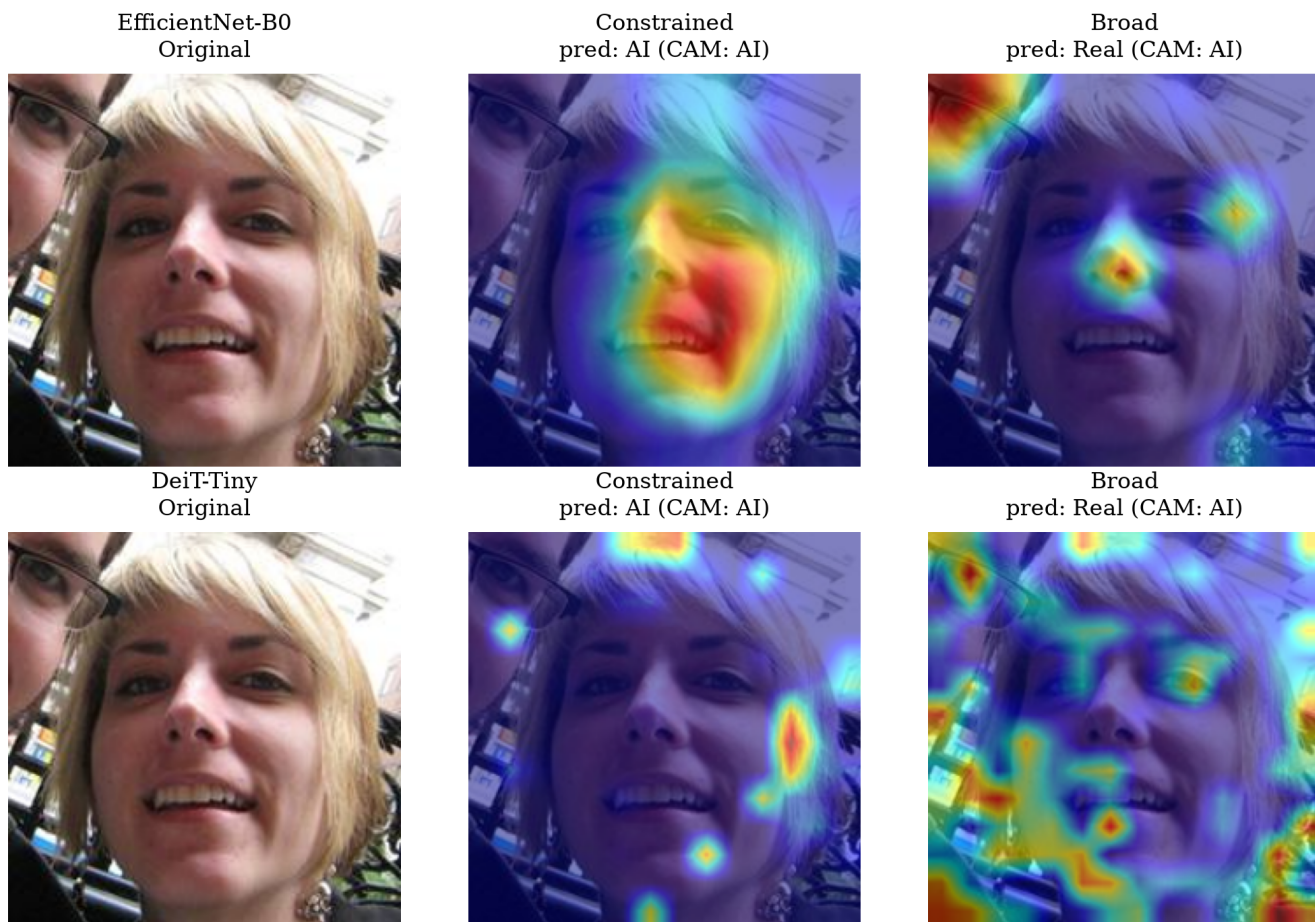


Figure 6. Control experiment: Grad-CAM on the same photo with the target class fixed to AI for all four models (constrained/broad $\times$ both architectures), regardless of each model’s actual prediction. The AI-evidence hotspot still relocates between training conditions, showing the shift in Figure 5 is not merely an artifact of the predicted class flipping.

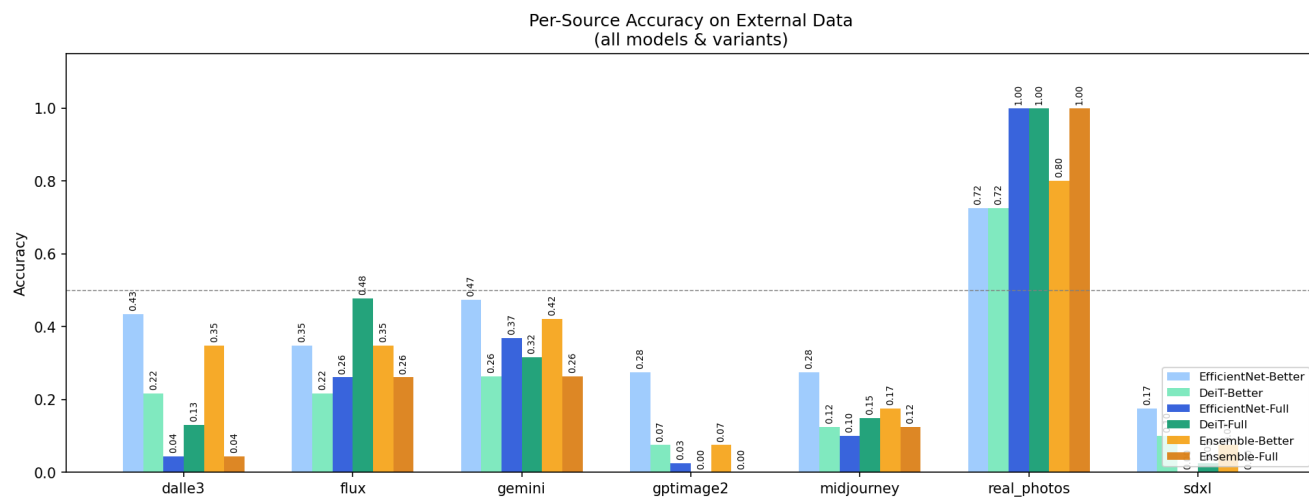


Figure 7. Per-source accuracy on external OOD data across all model variants.

ONNX Runtime Web is a viable platform for on-device detection: 16–22 MB models, sub-100 ms inference, no server required, and no data leaves the device.

Acknowledgments. Training and evaluation used the ArtiFact dataset (Kaggle). Model architectures were sourced from the `timm` library with ImageNet pretrained weights. ONNX export and browser inference used ONNX Runtime (Python and WebAssembly). Training was conducted on Google Colab (NVIDIA T4 GPU). Development assistance was provided by Claude (Anthropic). All training runs, evaluation, results analysis, and project decisions were performed and directed by the author.